

AX-7300 Hydroacoustic Profiler

Operator and Service Manual — firmware 5.2.0. Aldergrove Instruments Ltd., Richmond, British Columbia. Document reference AX-7300-OSM-520.

1. Overview

The AX-7300 Hydroacoustic Profiler is a self-contained hydroacoustic profiling instrument released in September 2024. Product status: Current — primary product line.

The instrument records backscatter across the water column and stores profiles internally for later retrieval. It is rated to a maximum operating depth of 1000 metres and is intended for deployment from a vessel, a fixed mooring, or an autonomous platform.

This manual covers the AX-7300 only. Procedures for the AX-7200 are similar in outline but differ in several specifics, and the two manuals must not be used interchangeably.

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2. Safety

Read this section before handling the instrument. The pressure housing stores significant energy when pressurised. Never attempt to open the housing while it is pressurised or immediately after recovery from depth.

- Vent the housing using the bleed screw before opening.

- Do not exceed the rated operating depth, including transient excursions during recovery.
- The battery pack contains lithium-ion cells. Do not puncture, crush, or expose to temperatures above the rated maximum.
- Disconnect power before servicing the connector.

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3. Specifications

Model	AX-7300
Maximum operating depth	1000 m
Operating temperature range	-10 °C to 50 °C
Battery endurance	26 hours continuous
Charge time	3.0 hours
Weight in air	3.8 kg
Sampling rate	400 kHz
Beam width	4.2 degrees
Internal storage	256 GB
Housing material	titanium Grade 5
Connector	SubConn MCBH12M

External power input	12–36 V DC
Firmware version	5.2.0
Warranty period	36 months
Calibration interval	18 months

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4. Installation

4.1 Mechanical mounting

Mount the instrument with the transducer face unobstructed and at least 300 mm from any reflective surface. Use the supplied clamp set; do not clamp directly onto the transducer housing.

4.2 Electrical connection

The instrument accepts 12–36 V DC through the SubConn MCBH12M connector. Apply a thin film of the supplied silicone grease to the connector pins before mating. Mate the connector fully until the locking sleeve seats.

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5. Operation

5.1 Pre-deployment checks

- Confirm firmware is at version 5.2.0 or later.
- Confirm the battery indicates a full charge; a full charge takes approximately 3.0 hours.
- Confirm the internal clock is synchronised to UTC.
- Confirm available storage is sufficient for the planned deployment duration.
- Inspect the O-ring seat for debris and replace the O-ring if in any doubt.

5.2 Deployment

At the configured sampling rate of 400 kHz, a fully charged instrument records continuously for approximately 26 hours. Reducing the ping rate extends endurance approximately linearly.

5.3 Recovery and data retrieval

Rinse the instrument in fresh water on recovery and allow it to dry before opening. Retrieve data over the wired interface; wireless retrieval is not supported for full-resolution profiles.

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6. Maintenance and calibration

Return the instrument for factory calibration every 18 months. Calibration outside this interval invalidates the accuracy specification and may invalidate warranty claims relating to measurement performance.

Inspect and replace the main O-ring at every deployment cycle. Inspect the anode annually and replace it when 50 per cent consumed. Clean biofouling from the transducer face with a soft brush and fresh water only — do not use solvents or abrasives.

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7. Troubleshooting

The instrument reports faults as numbered error codes on the status display and in the deployment log file.

E-119	Transducer calibration drift beyond tolerance. Re-run the bench calibration routine and, if the fault persists, return the head for factory recalibration.
E-311	Hydrophone array element fault. One or more of the sixteen array elements is not responding. The unit will continue in degraded mode at reduced resolution.
E-142	Pressure sensor reading outside the expected envelope for the reported depth. Inspect the port for biofouling.

If a fault persists after the action described above, record the code, the firmware version, and the deployment log, and contact technical support before redeploying the instrument.

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8. Warranty

The AX-7300 is supplied with a 36-month warranty from the date of delivery, covering defects in materials and workmanship. The warranty excludes consumables — O-rings, anodes, and the battery pack after twelve months — and excludes damage caused by operation outside the rated depth or temperature envelope.

Warranty claims require the instrument serial number and the deployment log covering the failure.

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9. Appendix — part numbers

AX7300-BC-SPARE-ORING	Main housing O-ring, pack of five
AX7300-BC-SPARE-ANODE	Sacrificial anode
AX7300-BC-BATT	Replacement battery pack
AX7300-BC-CLAMP	Mounting clamp set
AX7300-BC-CABLE-5M	Deck cable, 5 metre

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